



Wildfire Smoke Detection

In association with Whatcom Million Trees Project

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MAKE WAVES.

Abstract

This project is the implementation of a low-power wildfire smoke detection system intended for deployment in remote environments. The goal is to detect smoke-related air quality changes and wirelessly transmit alerts over long distances without requiring internet connectivity. The STM32 code was developed using STM32CubeIDE and integrates sensor communication, threshold-based smoke detection, low-power sleep functionality, and peer-to-peer LoRa messaging. During testing, significant effort was devoted to debugging UART and I2C communication interfaces. The software architecture was designed to support smoke detection, and alert transmission was implemented. Hardware communication issues were isolated through systematic debugging techniques, including UART terminal monitoring and I2C device scanning. The project demonstrates the feasibility of low-cost distributed wildfire monitoring systems and provides a foundation for future field-deployable sensor networks.

Background

Wildfires continue to increase in frequency and severity throughout many regions of North America. Early detection of smoke can significantly improve emergency response times and reduce environmental damage.

Challenges include:

- Remote deployment locations
- Limited access to cellular networks
- Low power requirements
- Long-range communication needs

LoRa technology provides long-distance wireless communication while maintaining low power consumption, making it suitable for wilderness monitoring applications.



References

1. Bosch Sensortec. BME68x Sensor API.
2. Reyax Technology. RYLR896 LoRa Module Datasheet.
3. STMicroelectronics STM32F030R8 Reference Manual.
4. STM32 HAL Driver Documentation.

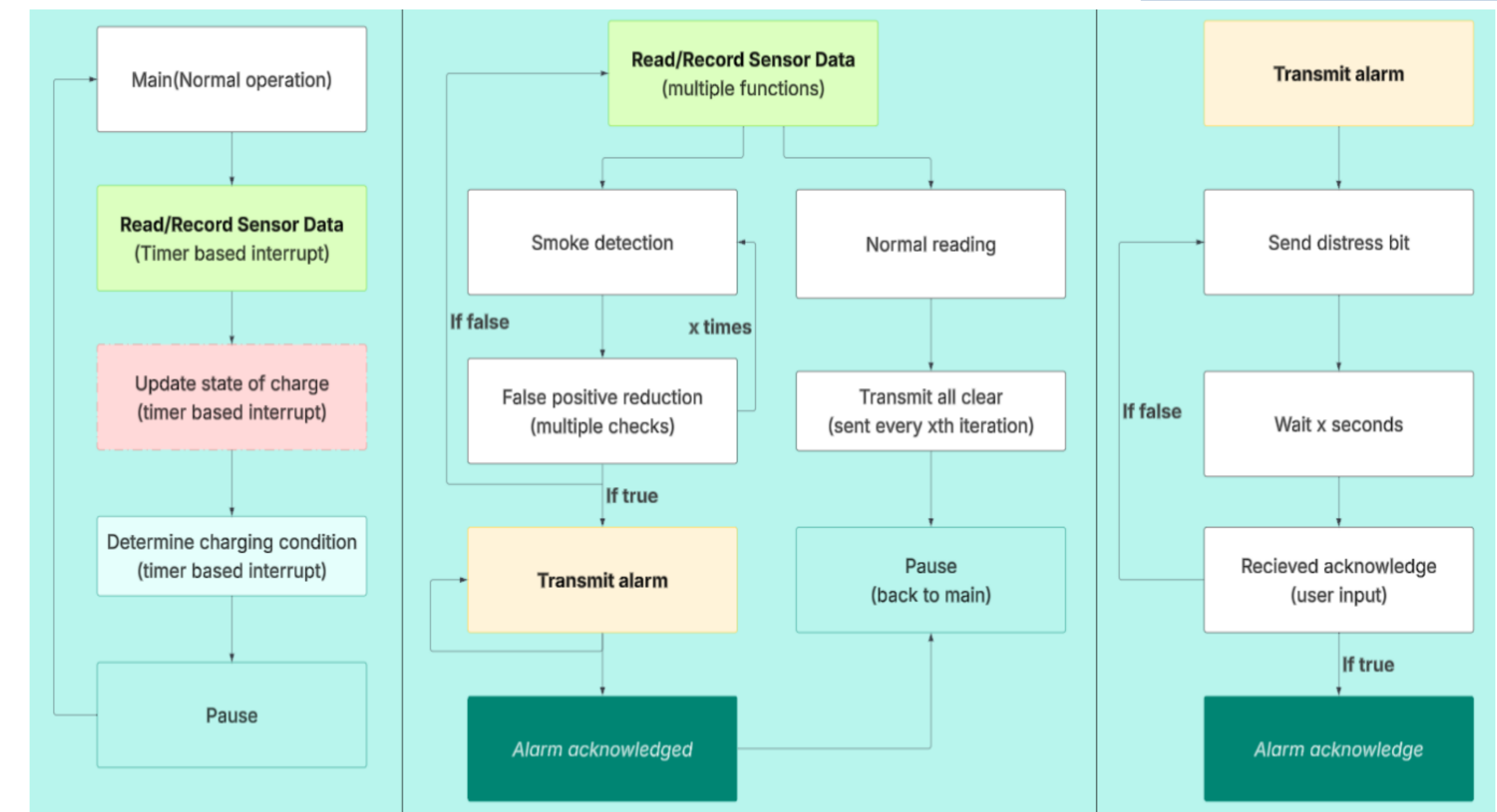
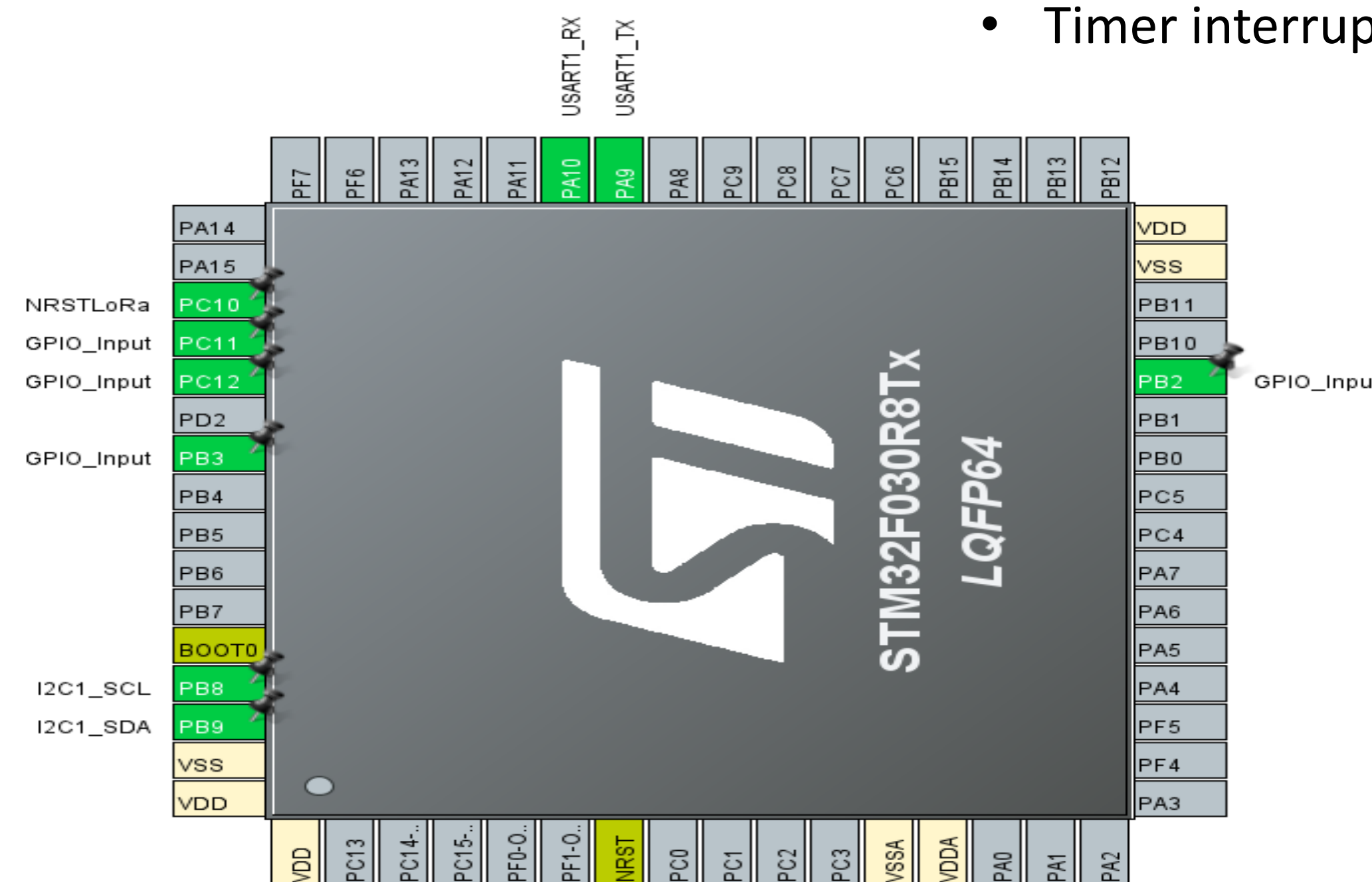
Methods and Materials

Hardware

- STM32 Nucleo-F030R8
- Bosch BME688 gas sensor
- Reyax RYLR896 LoRa module
- Custom PCB
- 3.3V power supply

Software

- STM32CubeIDE
- STM32 HAL drivers
- Bosch BME68x Sensor API
- UART communication
- I2C communication
- Timer interrupt scheduling



Results

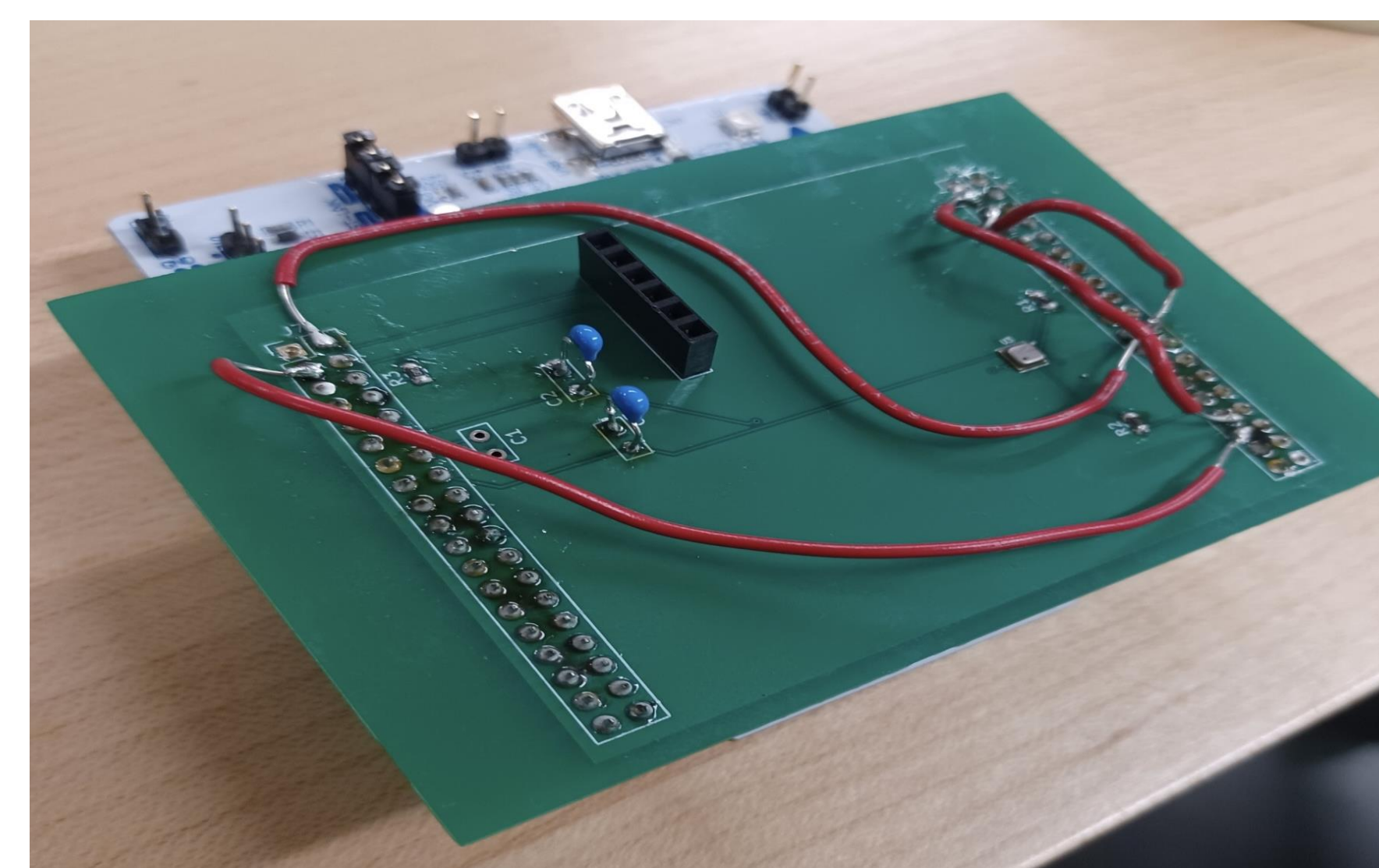
STM32 Firmware Execution	Pass
GPIO Verification	Pass
USART2 Debug Interface	Pass
COM4 Terminal Communication	Pass
LoRa Software Integration	Pass
BME688 Initialization	Failed
I2C Device Scan	No Device Detected
Reyax UART Communication	No Response Detected

Successful Outcomes

1. STM32 code development completed
2. UART debugging framework established
3. LoRa peer-to-peer architecture implemented
4. Smoke detection logic developed
5. Hardware testing procedures established

Issues Identified

1. BME688 sensor initialization failure
2. No detectable I2C response from sensor
3. Reyax module did not return AT command responses
4. Potential wiring issues identified during debugging



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HELLO FROM STM32
SIMULATED SMOKE EVENT
BME688 INIT FAILED
Starting I2C Scan...
I2C Scan Complete
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Future Direction

- Verify sensor and LoRa module wiring
- Complete hardware validation
- Implement field testing
- Deploy multiple sensor nodes
- Develop centralized monitoring station
- Improve smoke detection calibration
- Implement adaptive thresholding algorithms

Acknowledgements

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Capstone Project
Electrical & Computer Engineering Program